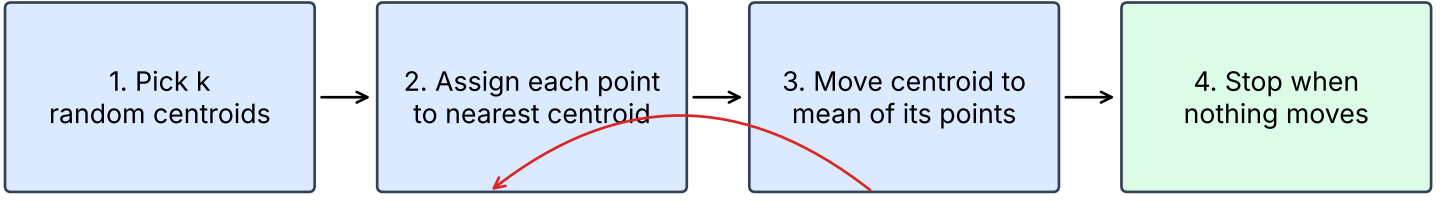The k-means loop